

YINING SHE

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EDUCATION

Carnegie Mellon University

Ph.D. in Software Engineering, Advised by Eunsuk Kang
Software and Societal Systems Department, School of Computer Science
Research focus: Trustworthy and Safe AI; Software Engineering for AI

Pittsburgh, PA
Aug.2022-Sep.2027(*expected*)

ShanghaiTech University

Bachelor of Engineering in Computer Science and Technology
GPA: **3.83/4.0**
Selected Honors: *Outstanding Graduate, Merit Student, Outstanding Teaching Assistant*

Shanghai, China
Sep.2018-Jun.2022

PROFESSIONAL SKILLS

Programming Languages

Python, C/C++, C#, R, MATLAB

Tools and Frameworks

PyTorch, scikit-learns, Tensorflow, OpenCV, Alloy, Git

DOCTORAL RESEARCH

Safeguarding LLM Agents from Misalignment through Provenance Analysis

In ASE 2026

- Proposed a provenance-based conceptual guardrail framework for detecting misaligned agent actions, which formalizes alignment checking as determining whether proposed tool calls are grounded by traceable evidence in the agent's context
- Categorized agent tool invocation misalignment into three types: tool-level, parameter-level, and interpretation-level misalignment
- Developed PROVENANCEGUARD, a multi-stage runtime-guardrail implementation of the framework that detects three types of misalignment step by step
- Evaluated PROVENANCEGUARD across 11 backbone LLMs, and results on a manually annotated benchmark show that the proposed method reduces average error rate on misaligned traces from 44.3% to 2.1% while not introducing a statistically significant increase in unnecessary interventions on aligned traces

FAIRSENSE: Long-Term Fairness Analysis of ML-enabled Systems

In ICSE 2025

- Proposed a simulation-based framework FAIRSENSE to detect and analyze the long-term unfairness in ML-enabled systems
- Modeled feedback loop between an ML-enabled system and the environment where system will be deployed, and designed a monte-carlo simulation framework to analyze the effect of feedback loop in long term
- Employed sensitivity analysis technique on simulation traces to identify the most impactful system configuration variables on long-term fairness
- Applied sampling heuristic to efficiently explore exponentially large configuration space without affecting sensitivity analysis results
- Evaluated FAIRSENSE on three real-world benchmarks: loan approval, opioid risk scoring, and predictive policing, and results show it can effectively detect long-term fairness issues and identify system configuration variables with the greatest impact for developers

INDUSTRY EXPERIENCE

Meta

Menlo Park, CA

ML Intern, AI Foundation Team, Central Product (Paper in submission)

May 2026-Jul.2026

- Developed an efficient evaluation feature “fast-mode” for the production evaluation framework of an internal agent that has >20k monthly active users and cost >\$6.6M/mo on evaluation
- This feature enables executing only around 60% of a benchmark with <1% error on the measured whole-benchmark pass rate, which projects to save \$30M/yr when applied to all evaluations
- Proposed three efficient evaluation methods on the recurring evaluation setup: historical caching, tinyBenchmark, and IRT-based adaptive testing
- Evaluated the three methods offline using the benchmark run data of 574 versions of the internal agent collected over 52 days, and turned the winner into the “fast-mode” feature

Oracle Cloud Infrastructure

GenAI Research Intern, Advised by Dan Roth (Paper in submission)

Redwood City, CA

May 2025-Dec.2025

- Conducted a systematic evaluation study of 5 LLM-based guardrail models when facing RAG-style contexts
- Curated a RAG-context input and output guardrail evaluation benchmark using 6.7k harmful prompts collected from 7 benchmarks and 8 generation LLMs
- Confirmed that inserting benign documents into the guardrail context alters the judgments of input and output guardrails in around 11% and 8% of cases
- Analyzed how retrieved content, user query, and LLM response affect guardrail's robustness separately, and explored how effective dedicated prompting and deep-reasoning model can mitigate the robustness gap

Intel Lab

AI Software Research Intern, Parallel Computing Lab

Hillsboro, OR

May 2024-Aug.2024

- Developed an ML-based tensor kernel optimization method to predict high-performing GEMM loop configurations across 15+ diverse machines
- Built a progressive ranked exploration pipeline that identified the globally optimal configuration on all tested machines while searching $\leq 3\%$ of the 1,040-configuration space

PUBLICATIONS

- Yining She, Yiliang Liang, Eunsuk Kang. “**Safeguarding LLM Agents from Misalignment through Provenance Analysis**”, In the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026.
- Yining Hong, Yining She, Eunsuk Kang, Christopher S Timperley, Christian Kästner. “**Don’t Make Models Guess Security and Safety: Symbolic Guardrails for Domain-Specific AI Agents**”, preprint on arXiv 2026.
- Ian Dardik, **Yining She**, Sam Procter, Keaton Hanna, Lutz Wrage, Eunsuk Kang. “**FASR: Automated Identification of Unsafe Control Actions in STPA**”, preprint on arXiv 2026.
- Guangyao Chen, Zeyu Li, Jicheng Gu, **Yining She**, Chenyang Zhu, Zhihao Jiang. “**Enhancing Context Awareness with Model Checking-based Uncertainty Representation in Decision Support Systems**”, In Formal Aspects of Computing 2026.
- Yining She, Daniel W Peterson, Marianne Menglin Liu, Vikas Upadhyay, Mohammad Hossein Chaghazardi, Eunsuk Kang, Dan Roth. “**RAG makes guardrails unsafe? Investigating robustness of guardrails under RAG-style contexts**”, preprint on arXiv 2025.
- Yining She, Sumon Biswas, Christian Kästner, Eunsuk Kang. “**Fairsense: Long-Term Fairness Analysis of ML-Enabled Systems**”, In the 47th IEEE/ACM International Conference on Software Engineering (ICSE) 2025.
- Sumon Biswas, Yining She, Eunsuk Kang. “**Towards Safe ML-based Systems in Presence of Feedback Loops**”, In SE4SafeML, Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) 2023.
- Mohammad Piran, **Yining She**, Renzhi Tang, Zhihao Jiang, Yash Vardhan Pant. “**Stable Interaction of Autonomous Vehicle Platoons with Human-Driven Vehicles**”, In American Control Conference (ACC) 2022.

PROFESSIONAL SERVICE

Program Committee

LLMTrust Workshop @ FSE (2026)

Reviewer

ICML (2026, **Gold Reviewer**), REALM Workshop @ EMNLP (2026)

TEACHING EXPERIENCE

Lead TA for graduate course “Software System Design”

Carnegie Mellon University, Spring 2026

TA for graduate course “Formal Methods”

Carnegie Mellon University, Fall 2023

Lead TA for undergraduate course “Algorithms and Data Structure”

ShanghaiTech University, Fall 2021

Lead TA for undergraduate course “Software Engineering”

ShanghaiTech University, Spring 2021

TA for undergraduate course “Algorithms and Data Structure”

ShanghaiTech University, Fall 2020